

Walkthrough: the Slosh Mechanism and the Parity Measurement

Q2.1 of the PI-question analyses, slowly, piece by piece
(companion to `pi_questions_analysis.ipynb`, §Q2.1)

2026-08-20

Setting: the $4\pi G = 6$, $\text{amp} = 0.2$ swing-collapse runs. The question being explained: how the collapse chooses its site, and how we established that the mirror symmetry of the problem is preserved by the code. Every claim below points at a measured curve in the figures.

1 The slosh mechanism: how the collapse chooses its site

Step 1 — what “the site” means

After fragmentation, one dominant knot forms on the corotation column $x \approx 0$. The mirror symmetry (§2) allows exactly two single-knot sites there: $y = 0$ (“center”) and $y = \pm\pi$ (“wrap”). The puzzle: from identical initial conditions, the 128^2 run picked center and the 256^2 runs picked the wrap. Which site wins, and why does it depend on resolution?

Step 2 — the collapse is not one-shot

Follow the gray max-density curve in Fig. 1. The swing compresses the crest to $\rho \approx 10$ –20 around $t \approx 4.5$ –5 — and then the crest *bounces*: pressure wins, and by $t = 6$ –8 the maximum is back down to $\rho \approx 2$. The famous “fragmentation at $t \approx 4.5$ ” is the beginning of a long negotiation, not the final collapse. The real runaway happens three to five time units later, out of the bounced debris.

Step 3 — what oscillates in the interlude

Take the corotation column’s density profile $\rho(y)$ and project it onto $\cos y$ and $\sin y$ (the $m=1$ Fourier coefficient c_1). Its magnitude $|c_1|$ says how much density contrast the column carries; its phase says *where* the pile-up is. Because of the mirror symmetry the phase can only be 0 or π (§2, Step 6): the pattern is a **standing wave** that alternately piles density at $y = 0$ (blue points in Fig. 1) and at $y = \pm\pi$ (orange points), passing through zero contrast at each flip.

Step 4 — why it oscillates instead of growing

By $t \approx 7$ the wave has rewound to $k_x = -6 + 1.5t \approx 4.5$, so $k = \sqrt{k_x^2 + k_y^2} \approx 4.6$, and the local dispersion relation gives

$$\omega^2 = \kappa^2 + c_s^2 k^2 - 4\pi G \rho_0 \approx 1 + 21 - 6 = +16 > 0 : \quad (1)$$

pressure has re-stabilized the winding wave — it *rings* instead of growing, with period $2\pi/\omega \approx 1.6$, i.e. a site flip every ≈ 0.8 . That matches the measured flip cadence in Fig. 1; and since $k(t)$ keeps

growing, the flips accelerate toward $t = 9$ – 10 , exactly as the 256^2 panel shows. (This is the same physics as the movie-flicker ringing of Q0.2, now with self-gravity in the frequency.)

Step 5 — the trigger

Meanwhile the bounced debris slowly re-contracts: the gray curve creeps from $\rho \approx 2$ up through ≈ 10 over $t = 7$ – 9 . A knot goes into runaway when its *local* Jeans length $\lambda_J \propto \rho^{-1/2}$ drops below its own size — after that, the gray curve goes vertical. Measured onset (last upward crossing of $\rho = 20$): $t = 8.80$ at 128^2 , $t = 9.45$ at 256^2 .

Step 6 — the selection

The runaway grabs the densest material *at the moment it fires* — and where that is alternates with the slosh. At 128^2 the onset lands on a $y=0$ phase (blue, $|c_1|$ risen to 1.5): clump at center, $(+0.025, -0.025)$. At 256^2 the onset comes half a slosh cycle later and settles on a π phase: knot at the wrap. Same mechanism, opposite outcomes, decided by a ≈ 0.65 timing offset.

Step 7 — why the resolutions fire at different times

Through the swing the two runs are essentially identical (d_{cos} agrees to 2–3 digits at $t \leq 5$). The *bounce*, however, compresses the crest to near the grid scale at 128^2 — this run exceeds its Truelove ceiling, the documented marginality — and that is precisely where resolution matters. The post-bounce trajectories drift apart, the 128^2 run tips over earlier, and the half-cycle offset flips the site. Not randomness: a deterministic, symmetric double-well decision balanced on a knife edge. The converged answer is the wrap: MG 256^2 , FFT 256^2 , and the AMR run (fine level $\equiv 256^2$) all agree.

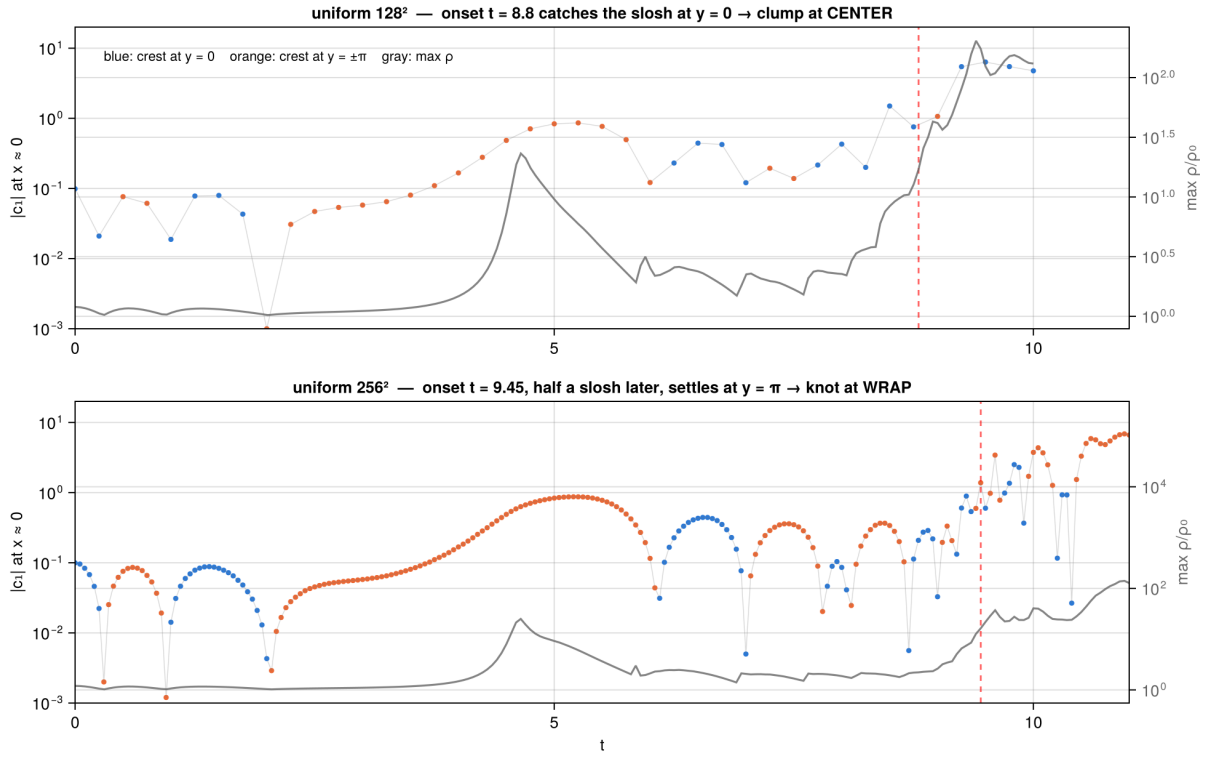


Figure 1: The slosh and the onset. Points: corotation-column $m=1$ amplitude $|c_1|$, colored by which site holds the density (blue $y=0$, orange $y = \pm \pi$); gray: max density; red dashed: runaway onset (ρ crosses 20).

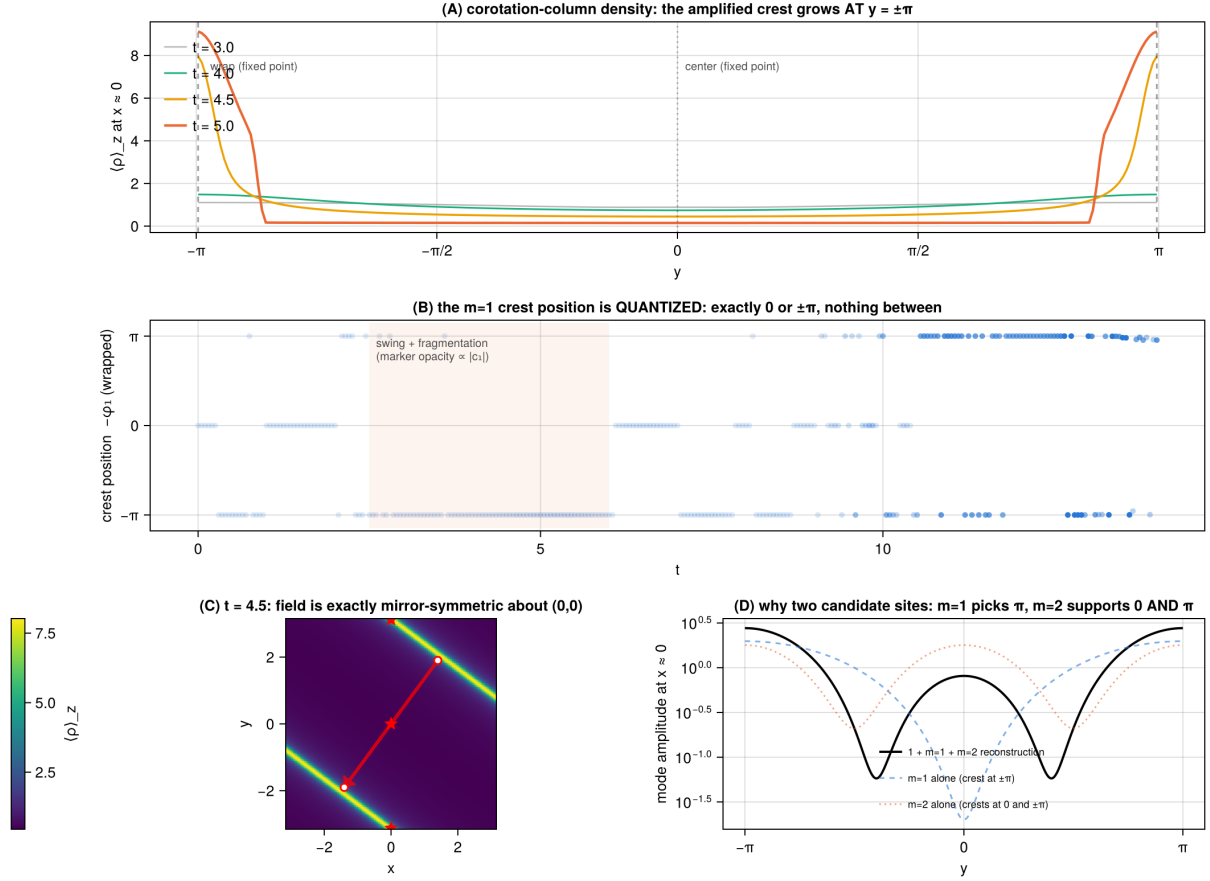


Figure 2: Why only two sites exist. (A) corotation-column profiles: the amplified crest grows at $y = \pm\pi$; (B) the measured crest position takes only the values 0 and $\pm\pi$; (C) the $t=4.5$ field with its mirror-symmetric structure (stars: fixed points; white dots: a mirror pair); (D) the $m=1 + m=2$ reconstruction of the column showing the two candidate sites.

2 The parity measurement

Step 1 — the symmetry being tested

The point reflection $P : (x, y) \rightarrow (-x, -y)$. The initial condition has it because cosine is even: $\cos(-k \cdot x) = \cos(k \cdot x)$. The *dynamics* keeps it because every term of the shearing-box equations transforms consistently under P (with velocities flipping sign): the background shear $-q\Omega x$ is odd in x , and Coriolis, tidal, pressure, and self-gravity terms all commute with the reflection. Crucially this holds at *any* amplitude — symmetry is not a linear-theory statement. So a P -even density field must stay P -even forever; any deviation is numerical.

Step 2 — the metric

Compare the field with its own reflection, snapshot by snapshot:

$$a(t) = \max_{i,j} |S(i, j) - S(\text{mirror}(i, j))| \quad (2)$$

on the z -averaged density S . $a = 0$ means the two mirror halves of the stored field are **bitwise identical** — all 32,768 mirror pairs, to the last bit.

Step 3 — the grid subtlety

Cell centers sit at $\pm\Delta/2, \pm3\Delta/2, \dots$; no cell lies at the origin. The mirror of cell index i is therefore $\text{mod1}(257 - i, 256)$, pairing $+\Delta/2$ with $-\Delta/2$. An off-by-one here (mirror = $258 - i$) misaligns every pair by one cell and reports a large *false* asymmetry — I made exactly that mistake on the first attempt. The self-check that catches it: the $t = 0$ initial condition must give $a = 0$ exactly.

Step 4 — reading the curve (Fig. 3)

Three regimes:

1. **The exact-zero rail**, through $t \approx 5.3$: $a = 0.0$, bitwise (138 of 281 MG snapshots; 145 for FFT). The entire swing and fragmentation onset preserve the mirror halves to the last bit.
2. **The ulp staircase**: snapshots are float32, whose smallest representable difference near a value ρ is one “ulp” $\approx 6 \times 10^{-8} \rho$. As the maximum density grows, that quantum grows with it — the measured $a(t)$ climbs in discrete steps that track the gray field maximum, while the *relative* asymmetry stays below 10^{-6} (invisible at storage precision) through $t = 12.70$ (MG) / 13.20 (FFT).
3. **The explosive break**, precisely when the runaway passes the Truelove ceiling: gravitational collapse amplifies any difference between the mirror halves — including pure round-off — at a rate that diverges as the collapse deepens. MG (knot $\rightarrow 2.6 \times 10^5$) shoots to $O(10^3)$; FFT (knot bounces near 700) breaks later and more gently. The difference maps show *where*: first one-ulp dust along steep gradient edges, then $O(1)$ asymmetry concentrated on the runaway knots.

Step 5 — what the measurement certifies

Any code defect that treats a location differently from its mirror — an off-center bias in the shear-periodic wrap, the FARGO shift, a ghost-zone exchange, or either gravity solver — would inject asymmetry at that location *every step*, to be amplified by the same instability that amplifies round-off. Staying at the storage floor for twelve time units, through a density contrast of $10^{2.5}$, means no such defect exists above the round-off level, in either solver. It also explains the “mirror-twin coin flips”: while the symmetry holds, the two twins of any off-axis structure are *exactly* degenerate, so the eventual winner is decided by the round-off asymmetry itself.

Step 6 — the bridge to the phase quantization

The same symmetry, viewed through the Fourier transform: for a mirror-even column profile, pairing y with $-y$ cancels every sin contribution, so the coefficients c_m are *real* — and a real number’s phase is 0 or π . That is why the measured crest position in Fig. 2(B) takes exactly two values and nothing in between: “the crest can only sit at center or wrap” and “the field is mirror-even” are one statement, written in two languages.

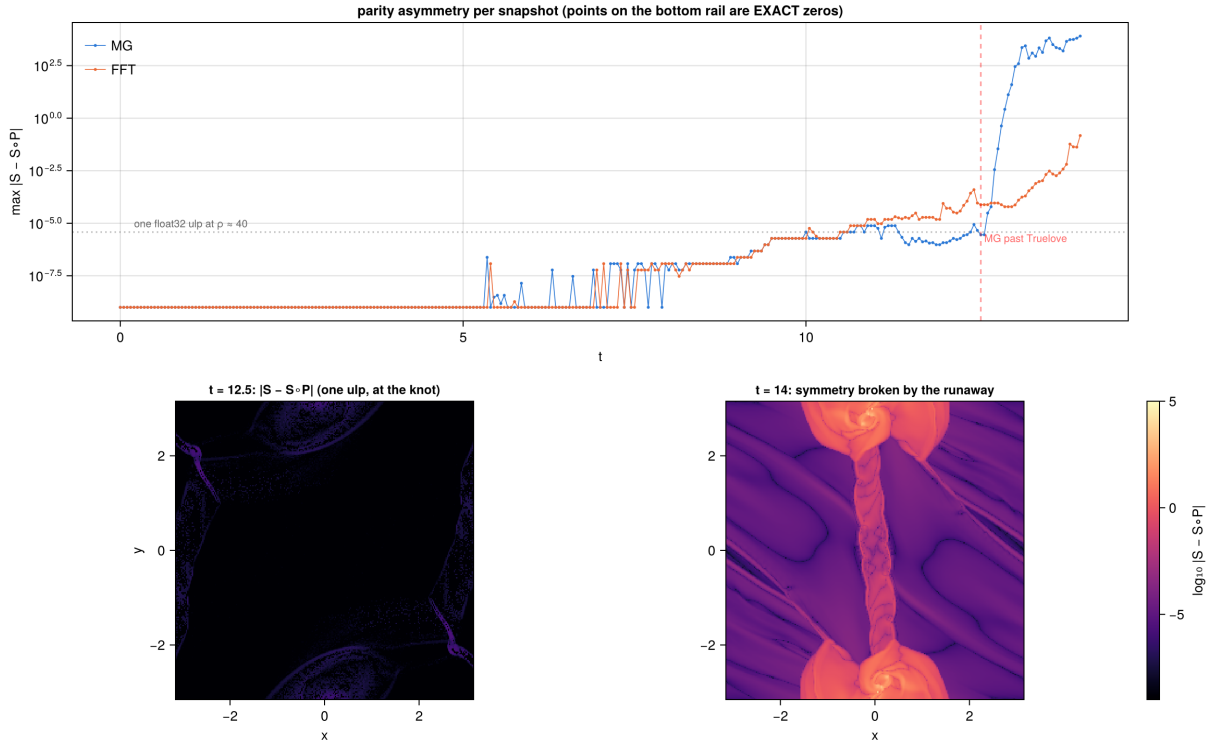


Figure 3: The parity measurement. Top: $a(t)$ for MG and FFT — exact-zero rail, ulp staircase, explosive break past the Truelove ceiling (red dashed). Bottom: $|S - S \circ P|$ maps at $t = 12.5$ (one ulp, tracing gradient edges) and $t = 14$ (symmetry broken at the runaway knots).